Reporting No: 7 Week No: 7 From :12 /09/2026 To:19/09/2026
Project ID: 31
Project Title: **CHARUSAT Online Course Assistant**

WEEKLY REPORT

Work done in last week (Attach supporting Documents):

- Improved the RAG retrieval pipeline by combining semantic search using FastEmbed with the BAAI/bge-small-en-v1.5 embedding model and keyword-based BM25 search.
- Implemented hybrid retrieval using Reciprocal Rank Fusion (RRF) to combine search results and improve the retrieval of relevant document chunks.
- Worked on document processing, including chunking, text overlap, metadata handling, and storing embeddings in ChromaDB.
- Reviewed the web-scraping and PDF-ingestion workflow used to collect official CHARUSAT programme information and prepare the knowledge base.
- Improved the chatbot's question suggestion feature using a verified question bank of 117 questions, allowing users to find relevant questions while typing.
- Reviewed the complete RAG architecture, including document ingestion, retrieval, LLM-based answer generation, and fallback mechanisms.

Reason for incomplete work:

- Further testing and optimization of retrieval accuracy are required for different types of user queries.
- Some improvements to the knowledge base and chatbot functionality require additional testing and validation.
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Plans for next week:

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- Perform further testing of the chatbot using different questions related to admissions, fees, eligibility, and university policies.
- Improve the accuracy and relevance of retrieved document chunks.
- Review the web-scraping and PDF-ingestion process to ensure the knowledge base contains clean and updated information.
- Test the frontend question suggestions and overall chatbot user experience.
- Document the project architecture, testing results, and improvements for the final project presentation.

References:

•CHARUSAT Online Education
https://www.charusat.ac.in/online-education

Signature of External Guide **Signature of Internal Guide**

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